

1. End-to-End CI/CD Pipeline ★★★★★

Tech: Jenkins + Docker + Kubernetes + GitHub + ArgoCD

Build:

Push code to GitHub

Jenkins builds Docker image

Push image to Docker Hub/ECR

Deploy automatically to Kubernetes

ArgoCD syncs deployment

Skills: CI/CD, GitOps, Docker, Kubernetes

2. Three-Tier Application Deployment ★★★★★

Tech: Nginx + Node/Java/PHP + MySQL + Docker + K8s

Features:

Frontend

Backend API

Database

Persistent Volumes

Ingress

Secrets & ConfigMaps

3. AWS Infrastructure using Terraform ★★★★★

Provision automatically:

VPC

Public/Private Subnets

NAT Gateway

EC2

Security Groups

Load Balancer

Auto Scaling

Everything in code.

4. Kubernetes Monitoring Stack ★★★★★

Install:

Prometheus

Grafana

Alertmanager

Node Exporter

Create dashboards:

CPU

Memory

Pods

Network

Disk

5. Centralized Logging Project ★★★★★

Deploy:

Elasticsearch

Logstash/Fluent Bit

Kibana

Collect logs from Kubernetes pods.

6. Shell Scripting Automation ★★★★★

Write scripts for:

Backup
Log Rotation
Disk Cleanup
Health Check
Service Monitoring
Email Alerts
User Management

7. Kubernetes Production Cluster ★★★★★

Deploy:

Nginx
Redis
MySQL
WordPress
Horizontal Pod Autoscaler
Rolling Updates
Liveness/Readiness Probes

8. GitHub Actions CI/CD ★★★★★

Instead of Jenkins:

Build
Test
Docker Build
Push Image
Deploy

9. Ansible Automation ★★★★★

Automate:

Package Installation
Docker Installation
User Creation
Nginx Configuration
Kubernetes Setup

10. Production Monitoring & Alerting ★★★★★

Monitor:

Linux servers
Disk usage
Memory
CPU
Services
Website uptime
Alerts via:

Email
Slack

Bonus Project (Excellent for Interviews)

DevOps Project: Netflix Clone Deployment

Pipeline:

GitHub

↓

Jenkins

↓

Docker Build



Docker Hub



Terraform



AWS EC2



Kubernetes



ArgoCD



Prometheus + Grafana



ELK Stack

This single project demonstrates almost every major DevOps skill.

If I were building a portfolio in 2026, I'd prioritize these 5:

- ✓ End-to-End CI/CD Pipeline (Jenkins + Kubernetes)
- ✓ Terraform AWS Infrastructure
- ✓ Kubernetes Monitoring (Prometheus + Grafana)
- ✓ ELK Logging Stack
- ✓ GitOps with ArgoCD

Production CI/CD Pipeline

Jenkins

SonarQube

Trivy

Docker

Kubernetes

ArgoCD

AWS Infrastructure as Code

Terraform

VPC

ALB

Auto Scaling

RDS

IAM

Route 53

Kubernetes Production Cluster

Ingress

HPA

Network Policies

Secrets

ConfigMaps

Persistent Volumes

Monitoring & Alerting

Prometheus
Grafana
Alertmanager
Custom alerts
Centralized Logging
EFK/ELK or Loki + Grafana
Log collection and analysis
Configuration Management
Ansible playbooks
Server provisioning
Application deployment
Container Security
Trivy
Image scanning
Secrets management
RBAC
Pod Security
Real-world Capstone
Deploy a microservices application with:
CI/CD
GitOps
Monitoring
Logging
Terraform-managed infrastructure
Automated deployment and rollback